TAI NGUYEN XUAN

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EDUCATION

- **University of Science and Technology of Hanoi (USTH)** *Oct 2022 – Aug 2026*
  Bachelor in Advanced Materials Science and Nanotechnology (AMSN)
  - GPA: 17.01/20.00 | Rank 1st in major (Valedictorian)
  - Thesis: "Fabrication and Characterization of Polydopamine-based Crossbar Devices"

RESEARCH EXPERIENCE

Laboratory of Advanced Materials and Application
University of Science and Technology of Hanoi

- ***Fabrication and Characterization of Polydopamine-based Crossbar Devices*** *Sep 2025 – Present*
  Research Assistant | Supervisor: Dr. Nguyen Van Quyen
  - Fabricated polydopamine (PDA) crossbar memory junctions by integrating electrodeposited thin films with electron-beam evaporated metal electrodes.
  - Performed DC I–V and temperature-dependent electrical characterization to investigate switching behavior and charge transport, supported by AFM, Raman, and UV–Vis analysis.
  - Developed LabVIEW-based automated measurement routines for fast-pulse switching, endurance, retention, and switching dynamics.

  ○ ***Visiting Research Scholar*** *June 2026*
  Molecular Electronics Lab, Guangdong Technion Israel Institute of Technology, China
  - Quantified short-term plasticity (STP), paired-pulse facilitation (PPF), and endurance to evaluate the temporal response and cycling stability of volatile PDA memristors.
  - Conducted XPS analysis of PDA thin films to examine surface composition and chemical states relevant to protonation and interfacial transport.

- ***Design and Fabrication of a Low-Temperature Cryostat System*** *Nov 2025 – Mar 2026*
  Project Group Leader | Supervisor: Dr. Nguyen Van Quyen
  - Designed and integrated a modular low-temperature electrical characterization system with cryogenic-compatible components and custom PCBs.
  - Modeled transient heat transfer using Python-based finite-difference simulations to guide thermal design, fabrication, and system integration.

- ***Simulation of Plasmonic Structures*** *Sep 2024 – Jun 2025*
  Research Assistant | Supervisors: Dr. Hoang Thi Hong Cam, Assoc. Prof. Nguyen Van Quynh
  - Modeled Au/Au@Ag plasmonic nanosphere arrays using Lumerical FDTD and HEAT to investigate optical resonance and associated thermal transport.

HONORS AND AWARDS

- **Merit Scholarship, USTH** *2023–2025*
- **Shaping the Future Initiative Scholarship, USTH** *2025–2026*
- **Rector's Award of Excellence, USTH** *2022–2025*
- **Internship Scholarship, USTH** *2025–2026*

PUBLICATIONS & MANUSCRIPTS

- **Nguyen, T. X.**, Le, T., Tran, V. T., Nguyen, Q., "Electric Field-Driven Interfacial Proton-Coupled Electron Transfer in Protonated Polydopamine Junctions." *Submitted to Journal of the American Chemical Society*, 2026.

ADDITIONAL TRAINING

- **Photovoltaic Summer School** *June 2025*
  Department of Electrical Engineering, Tatung University, Taiwan
  - Gained practical understanding of photovoltaic microgrid design, AIoT-based smart-energy systems, and carbon-footprint verification.
  - Participated in cultural exchange with students and professionals from ASEAN countries.

SKILLS

- **Fabrication:** Photolithography, e-beam evaporation, electrodeposition, thin-film processing
- **Materials Characterization:** XPS, AFM, Raman/SERS, UV–Vis, FTIR, optical microscopy
- **Electrical Characterization:** DC I–V, temperature-dependent I–V, fast-pulse characterization, endurance/retention
- **Instrumentation & Automation:** LabVIEW, instrument control, automated measurement workflows
- **Simulation & Engineering Design:** Lumerical FDTD/HEAT, COMSOL, finite-difference modeling, FreeCAD, KiCad
- **Programming & Research Computing:** Python (NumPy, pandas, Matplotlib), Git/GitHub, reproducible scientific workflows
- **Scientific Documentation:** Typst, LaTeX
- **Languages:** Vietnamese (native), English (IELTS 7.0), French (B1)

TEACHING & LEADERSHIP EXPERIENCE

- **Academic Teaching & Mentoring** *Sep 2023 – Mar 2026*
  Tutor & Peer Mentor
  - Tutored undergraduate students in fundamental physics and chemistry, providing structured academic support.

- **USTH Science Art Club** *Jul 2024 – Feb 2025*
  Club President
  - Led STEM outreach and taught physics and chemistry to visiting high-school students.
  - Trained members in scientific literature analysis and coordinated STEM exhibition projects.

- **USTH Student Association** *Mar 2023 – Jun 2024*
  Marketing Committee Member
  - Coordinated campus-wide student initiatives and communication with university offices.
  - Delivered a time-management workshop for incoming students.